Item 3 — Bottom endmembers: 6 types to 4 types

Four groups by Euclidean distance, one representative per group | 2026-09-03

Problem

The inversion uses four bands (443, 490, 560, 665 nm). Using six bottom types at once leaves too many unknowns. Coral, brown algae, red algae, and green algae are all dark, so they are hard to tell apart. The comment was to start with only 3–4 types that differ clearly in brightness.

Aim

Merge the six groups already defined in step 4 into four groups using Euclidean distance, and pick one representative in each group. These four types become the endmembers for the mixed-bottom inversion.

Method

Read reflectance at 443, 490, 560, and 665 nm. From these four values form a 7-D feature vector f, and measure similarity with Euclidean distance d(x, y) (equations below).

To form four groups, compute d among the six group-mean feature vectors (n = 7; Table 1, Figure 1) and merge nearby groups. To pick a representative, compute the 4-band mean (centroid) of the members and take the original six-group representative with the smallest 4-band distance (n = 4) to that centroid. This keeps the endmembers consistent with the HydroLight runs already used.

$$f = [\bar{R},\ R_{490}/R_{443},\ R_{560}/R_{490},\ R_{665}/R_{560},\ R_{490} - R_{443},\ R_{560} - R_{490},\ R_{665} - R_{560}]$$

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

$\bar{R}$ is the 4-band mean reflectance. Use n = 7 when merging groups and n = 4 when choosing a representative.

Distances among the six groups are given in Table 1 and Figure 1. The closest pair is G3–G4 (coral–dark vegetation, 0.29), then G1–G2 (sand–sediment, 0.36). Green algae (G6) is at least 1.58 from every other group. From this, dark vegetation and coral (G3+G4+G5) were merged, green algae was kept alone, and bright sand and sediment were kept as two groups because they occupy different brightness levels. That yields four groups.

Table 1. Euclidean distance between the six group-mean feature vectors (smaller = more similar)

	G1 Bright sand	G2 Medium sediment	G3 Coral / turf	G4 Dark vegetation	G5 Red algae	G6 Green algae
G1 Bright sand	—					
G2 Medium sediment	0.36	—				
G3 Coral / turf	0.93	0.61	—			
G4 Dark vegetation	1.10	0.77	0.29	—		
G5 Red algae	0.95	1.11	1.51	1.65	—	
G6 Green algae	2.50	2.23	1.65	1.58	2.93	—

Note: Yellow cells are the two closest pairs (G3–G4, G1–G2). The distance is d(x, y) above with n = 7.

Figure 1. Same distances as Table 1. Yellow is close, dark blue is far. G3–G4 (0.29) and G1–G2 (0.36) are close; green algae (G6) is far from all groups.

Results

The four groups and representatives are listed in Table 2. The four representatives are ooid sand, dark sediment, brown algae, and green algae. Their 4-band mean reflectances are 0.55, 0.24, 0.04, and 0.12, so brightness is separated, and only green algae has a 560 nm peak (Figure 2). In the dark group, among the original six representatives, brown algae (0.059) was closer to the centroid than coral (0.078) or red algae (0.074).

Table 2. Four groups and representatives (distance = 4-band Euclidean distance)

Group	Members merged	Representative	Distance to centroid
Bright sand	ooid sand, coral sand	ooid sand	0.117
Medium sediment	hardpan, dark sediment	dark sediment	0.088
Dark vegetation / coral	coral, turf algae, kelp, brown algae, red algae, seagrass, etc.	brown algae	0.059
Green algae	green algae	green algae	0.000

Note: For two-member groups the two distances are equal, so the existing representative is kept.

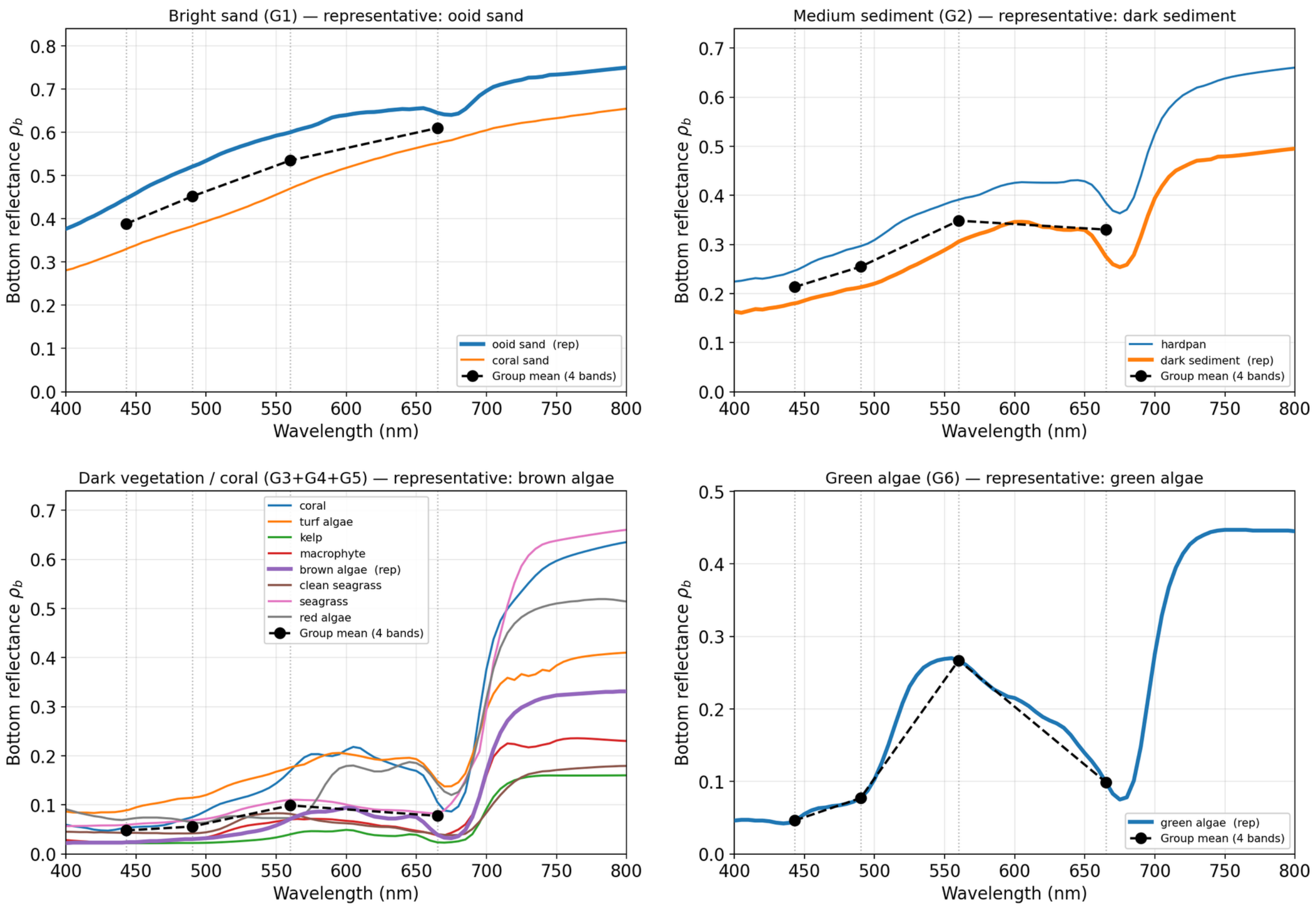

Figure 2. Spectra of the four groups. X-axis: wavelength (nm); Y-axis: bottom reflectance ρ_b. Thick line = representative; thin lines = other members; black dashed line with dots = 4-band group mean. Top left: bright sand; top right: sediment; bottom left: dark vegetation/coral; bottom right: green algae.

Conclusion

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The mixed-bottom inversion uses four endmembers: ooid sand, dark sediment, brown algae, and green algae. Coral, red algae, and the rest will be added later.